

**California Regional Water Quality Control Board****Central Valley Region****Robert Schneider, Chair**

C. Lloyd, Ph.D.
Secretary for
Environmental
Protection

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**Old
Business**

15 November 2005

Ms. Sherri Lucy
Del Rio East Home Owners Association
906 13th Street
Modesto, CA 95354

NOV 15 2005

NOTICE OF APPLICABILITY

WATER QUALITY ORDER NO. 97-10-DWQ-R5042, GENERAL WASTE DISCHARGE REQUIREMENTS FOR DISCHARGES TO LAND BY SMALL DOMESTIC WASTEWATER TREATMENT SYSTEMS, DEL RIO EAST HOME OWNERS ASSOCIATION, ONSITE WASTEWATER TREATMENT SYSTEM, STANISLAUS COUNTY

We reviewed the Report of Waste Discharge (RWD) in support of an ongoing discharge of up to 15,000 gallons per day (gpd) of domestic wastewater from an onsite wastewater treatment system (OWTS) owned by Del Rio East Homeowners Association (hereafter Discharger) serving the Del Rio East community. The RWD consists of a complete Form 200; technical submittals dated 13 May, 20 June, 2 August, and 20 September 2005; and a \$1,800 filing fee. Because the subject domestic discharge is less than 50,000 gpd, it is eligible for a 50 percent discount of the initial filing fee and subsequent annual fees. Staff intends to process a refund of a portion of the filing fee to reflect the discount as soon as feasible. The RWD contains all the technical information we require; therefore, the RWD is complete.

Based on the information provided in the RWD, we have determined that the discharge meets the conditions for approval under Water Quality Order No. 97-10-DWQ, *General Waste Discharge Requirements for Discharges to Land by Small Domestic Wastewater Treatment Systems* (hereafter General Order). All the requirements contained within the General Order described as applicable to "All Small Domestic Systems," "Activated Sludge Systems," and "Subsurface Disposal Systems" apply to your OWTS. You are hereby assigned General Order No. 97-10-DWQ-R5042.

PROJECT LOCATION

The OWTS serves 55 homeowners within the community of Del Rio East located along the Stanislaus River, just north of the City of Modesto and about 3.5 miles west of the City of Riverbank within the NE 1/4 of Section 20, T1S, R9E, as shown on Attachment A, which is attached hereto and made part of the Notice of Applicability (NOA) by reference. Surface water drainage is to the Stanislaus River.

The OWTS is in the San Joaquin River Basin. *Water Quality Control Plan for the Sacramento River Basin and San Joaquin River Basin, Fourth Edition* (hereafter Basin Plan), designates beneficial uses,

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establishes water quality objectives, and contains implementation programs and policies to achieve water quality objectives for all waters of the Basin.

The beneficial uses of the Stanislaus River, as identified in Table II-1 (water body #90) of the Basin Plan, include agricultural supply; water contact recreation; non-contact water recreation; wildlife habitat; and groundwater recharge. The beneficial uses of area groundwater include municipal and domestic supply; and agricultural supply.

The Regional Board's policy regarding the design and operation of onsite wastewater treatment systems is described in the enclosed *Guidelines for Waste Disposal from Land Developments* (hereafter Guidelines), which is incorporated in the Basin Plan.

PROJECT DESCRIPTION

The OWTS is owned and operated by the Discharger. The OWTS will replace an existing OWTS, which is deteriorating, with an activated sludge plant, manufactured by P. J. Hannah. The OWTS is sized for 25,000 gpd, but will be operated at flows not to exceed 15,000 gpd in order to incorporate denitrification.

Based on County records, the RWD reports that percolation rates determined during initial OTWS construction ranged from less than 1 to 2 minutes per inch. The RWD states that during the construction of the OWTS, a significant amount of fill materials ranging from clay loam to sandy loam was placed within the leachfield area. The RWD indicates that a 12-foot-deep test pit was excavated in 2005 to determine that the existing leachfield lines were installed 3 feet below ground surface (bgs), an intermittent sand lens exists 5-6 feet bgs, and groundwater was not encountered in the test pit. The RWD indicates that area groundwater is between 40 - 50 feet below ground surface (bgs), according to the *Lines of Equal Elevation of Water in Wells in Unconfined Aquifer*, published by Department of Water Resources (DWR) in Spring of 2000.

Regional Board staff letter, dated 23 August 2005, indicated that the rapid percolation rates cited in the RWD exceed that identified in the Guidelines for conventional onsite systems. Because the discharge, as originally proposed, threatened to pollute groundwater, the letter required the Discharger to revise its discharge (e.g., decreasing leachfield percolation rates by installing engineered fill or upgrading the new package plant to implement filtration and disinfection). The Discharger submitted a supplemental RWD, dated 20 September 2005, which proposed the latest OTWS design by incorporating filtration and disinfection.

The OWTS will consist of: a comminuter, aeration basin, clarifier, filtration unit, an ultraviolet (UV) disinfection system, a sludge holding tank and 1,314 linear feet of existing leachfield lines. To protect the beneficial uses of groundwater, effluent discharged to the leachfields will be characterized by 20 mg/L biochemical oxygen demand (BOD), 20 mg/L total suspended solids (TSS), 10 mg/L total nitrogen, and 23 MPN/100 mL total coliform organisms (TCO). The effluent limits set forth in this NOA are based on the effluent quality characterized in the Discharger's RWD.

The source water for the Del Rio East community is from two onsite groundwater wells approximately 1,500 feet south of the OTWS. The nearest precipitation gage for which data is readily available is a

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weather station in the City of Modesto, operated by the Modesto Irrigation District, which reports an average annual rainfall of about 8 inches.

This NOA regulates the OWTS and its discharge of domestic wastewater to land.

FACILITY-SPECIFIC REQUIREMENTS

1. Discharge of wastewater at a location or in a manner different from that described in the RWD is prohibited.
2. The existing OWTS shall be operated as described in the RWD and in accordance with the requirements contained in the General Order and in the enclosed *Guidelines for Waste Disposal from Land Developments*, whichever are more stringent.
3. Influent flow to the OWTS shall not exceed 15,000 gpd.
4. Prior to initiating discharge to the OWTS, the Discharger shall submit a technical report certifying that the OWTS was constructed in accordance with the RWD and is capable of complying with the General Order and this NOA. The technical report is subject to the requirements of Facility-Specific Requirement 7 and is subject to Executive Officer written approval.
5. Effluent discharged to the leachfield shall not exceed the following limits:

	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>
BOD	mg/L	20	40
TSS	mg/L	20	40
Total Nitrogen	mg/L	10	--
Total Coliform Organism	MPN ¹ /100 mL	23	240

¹ Most probable number

6. As a means for determining compliance with General Order Requirement B.3, the Discharger shall include in each annual report (due 15 January) a copy of a log documenting the quantity and method of disposal of all solids (e.g., screenings and sludge) removed from the OWTS during the previous calendar year.
7. All technical reports required herein that involve evaluation, or other work requiring interpretation and proper application of engineering or geologic sciences, shall be prepared by or under the direction of persons registered to practice in California pursuant to California Business and Professions Code, sections 6735, 7835, and 7835.1. To demonstrate compliance with Title 16, California Code of Regulations, sections 415 and 3065, all technical reports must contain a statement of the qualifications of the responsible registered professional(s). As required by these laws, completed technical reports must bear the signature(s) and seal(s) of the registered professional(s) in a manner such that all work can be clearly attributed to the professional responsible for the work.
8. The Discharger shall submit the required annual fee (as specified in the annual billing issued by the State Water Resources Control Board) until the NOA is officially terminated.

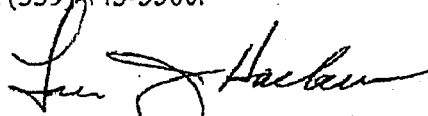
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9. Failure to abide by the conditions of the General Order and this letter authorizing applicability, including its supplemental monitoring and reporting requirements, could result in enforcement actions as authorized by provisions of the California Water Code.

If you have any questions regarding this NOA or associated fees, contact Ms. Alexis R. Phillips-Dowell at (559) 445-5500.

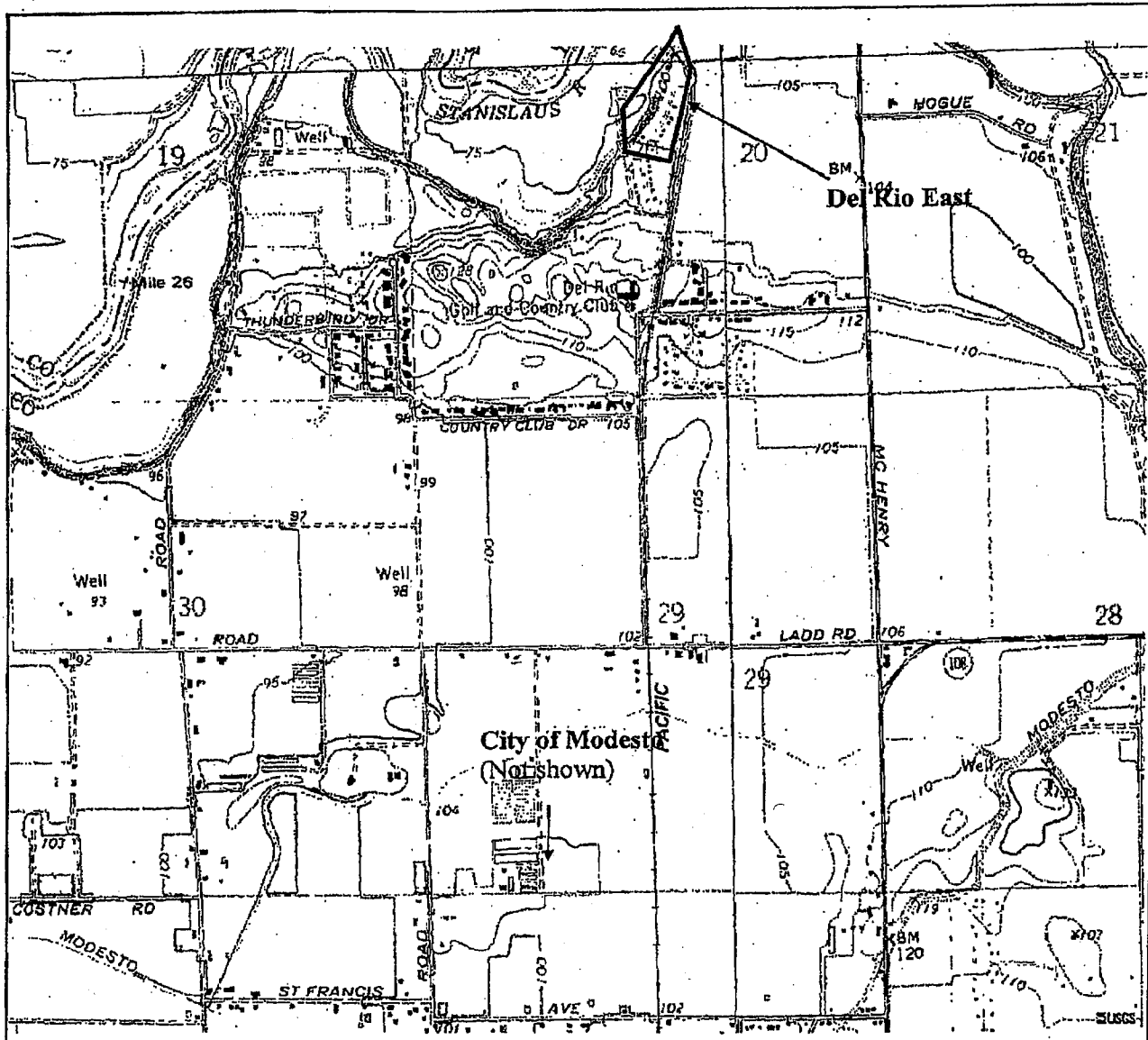

THOMAS R. PINKOS
Executive Officer

Enclosures: Attachment A - Location Map
General Order:

Monitoring and Reporting Program No. 97-10-DWQ
Standard Provisions and Reporting for Waste Discharge Requirements
Guidelines for Waste Disposal from Land Developments

cc w/out enclosures: Mr. Denise Wood, Stanislaus County Environmental Resources Department,
Modesto
Mr. Dan Hinrichs, DJH Engineering, Modesto
Mr. Ralph Becker, 7309 Del Cielo Way, Modesto, CA 95356

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ATTACHMENT A

LOCATION MAP

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DEL RIO EAST HOME OWNERS ASSOCIATION
ONSITE WASTEWATER TREATMENT SYSTEM
STANISLAUS COUNTY



Not to Scale